

Networked Control Systems: A Survey of Trends and Techniques

Xian-Ming Zhang, *Senior Member, IEEE*, Qing-Long Han, *Fellow, IEEE*, Xiaohua Ge, *Member, IEEE*, Derui Ding, *Member, IEEE*, Lei Ding, *Member, IEEE*, Dong Yue, *Senior Member, IEEE*, and Chen Peng, *Senior Member, IEEE*

Abstract—Networked control systems are spatially distributed systems in which the communication between sensors, actuators, and controllers occurs through a shared band-limited digital communication network. Several advantages of the network architectures include reduced system wiring, plug and play devices, increased system agility, and ease of system diagnosis and maintenance. Consequently, networked control is the current trend for industrial automation and has ever-increasing applications in a wide range of areas, such as smart grids, manufacturing systems, process control, automobiles, automated highway systems, and unmanned aerial vehicles. The modelling, analysis, and control of networked control systems have received considerable attention in the last two decades. The ‘control over networks’ is one of the key research directions for networked control systems. This paper aims at presenting a survey of trends and techniques in networked control systems from the perspective of ‘control over networks’, providing a snapshot of five control issues: sampled-data control, quantization control, networked control, event-triggered control, and security control. Some challenging issues are suggested to direct the future research.

Index Terms—Event-triggered control, networked control systems, quantization control, sampled-data control, security control.

I. INTRODUCTION

THE rapid development of network technologies has brought much change to people’s life. Modern communication networks can provide swift and reliable communication between any two or more physical plants located in different places. Such prominent characteristics make communication networks extensively used to connect control components within a control loop, leading to the so-called networked control systems (NCSs). NCSs have been applied in various areas, such as space exploration, environments, industrial

automation, robots, aircraft, automobiles, manufacturing plant monitoring, remote diagnostics and troubleshooting, and tele-operations [1]–[6].

An NCS inherits both advantages and disadvantages from communication networks. On the one hand, communication networks allow data packets to be shared among control components, which means that some traditional point-to-point wiring in the installation of a control system may be avoided. As a result, control cost incurred from installation and maintenance can be reduced significantly. Besides, the well-used communication protocols can ensure data packets to be successfully transmitted between control components, which renders an NCS of high reliability. On the other hand, communication networks transmit data packets only in the form of digital signals rather than continuous signals. This means that signals from physical plants should be sampled and quantized before being transmitted. Moreover, since the network bandwidth is limited, network traffic congestion is unavoidable, usually leading to some constraints such as network-induced delays and packet dropouts. Notice that signal sampling, quantization errors, network-induced delays and packet dropouts may have the effects on control performance of an NCS. Therefore, how to mitigate the effects of those network imperfections on control performance of NCSs has received considerable attention, see, e.g., survey papers [4], [7]–[10] and references therein.

Two of the key research directions for NCSs are ‘control of networks’ and ‘control over networks’ [11]. ‘Control of networks’ aims at providing pretty good quality of service (QoS) of communication networks such that NCSs can achieve satisfactory control performance. In this scenario, network data flows perform very well and network resources can be utilized efficiently and fairly. ‘Control over networks’ targets at devising suitable control strategies to reduce the effects of network imperfections on desired control performance. Clearly, ‘control of networks’ focuses on how to improve the QoS of networks while ‘control over networks’ on how to enhance control robustness against network constraints. The former is within the field of computer science while the latter falls into the scope of control systems. In this paper, we place emphasis on the latter, i.e., ‘control over networks’.

Recalling the existing literature, a rather large number of results have been reported on ‘control over networks’. Among those results, it is found that more attention is paid to the following five issues:

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X.-M. Zhang, Q.-L. Han, X. Ge, and D. Ding are with the School of Software and Electrical Engineering, Swinburne University of Technology, Melbourne, VIC 3122, Australia (e-mail: xianmingzhang@swin.edu.au; qhan@swin.edu.au; xge@swin.edu.au; dding@swin.edu.au).

L. Ding and D. Yue are with the Institute of Advanced Technology, Nanjing University of Posts and Telecommunications, Nanjing 210023, China (e-mail: dl522@163.com; medongy@vip.163.com).

C. Peng is with the Shanghai Key Laboratory of Power Station Automation Technology, School of Mechatronic Engineering and Automation, Shanghai University, Shanghai 200072, China (e-mail: c.peng@shu.edu.cn).

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Sampled-data control, which is to investigate how sampling affects control performance of a control system. Since digital techniques are widely implemented in real-world control systems of continuous-time dynamics, it is more practical that the controller is designed based on sampled data. In a sampled-data control system, one of the fundamental issues to be addressed is how to select a proper sampling period because it directly affects stability of the closed-loop control systems.

Quantization control, which is to study the quantization effects on control systems. The process of converting signals from analog to digital is an example of quantization, which maps input values from a larger set to output values in a smaller set. Quantization can be regarded as a lossy compression procedure. The resultant quantization errors between an input value and its quantized value definitely may deteriorate system performance or destroy system stability.

Networked control, which admits network-induced delays and packet dropouts to exist. Its objective is to investigate the effects of network-induced delays and packet dropouts on control systems. For example, how large is the network-induced delay (or the number of consecutive dropouts) endurable for a control system to retain certain performance? The core issue on such investigations is how to establish a relevant mathematical model for those network-induced constraints. By employing a time-delay system model, an interesting finding is that proper network-induced delays possibly play a *positive* role in improving system performance [13].

Event-triggered control, which has become a hot topic since 2007. Under an event-triggered sampling scheme, whether or not to sample system signals is determined by an event rather than the elapse of a fixed period [14]. As a result, computation resources, energy resources and network resources of an NCS can be saved significantly. However, when external disturbances are imposed on the physical plant, the positive minimum inter-event time between any two consecutive sampling instants may not be ensured [15], leading to the so-called Zeno behaviour. In order to elude such Zeno behaviour, a sampled-data-based event-triggered transmission scheme is proposed in [16], which allows system signals to be sampled periodically while an event-generator is used to determine if the sampled signals should be released to the communication network.

Security control, which addresses the security of an NCS in the presence of malicious attacks. Although communication networks provide a convenient platform for wide applications of modern control systems, they are currently vulnerable to malicious attacks due to their openness and insufficient protection. For example, by injecting malicious computer malware, hackers may falsify data packets or change the transmission paths of data packets; and by stealing the encryption key, cyber attackers may illegally get access to the remote control center, resulting in collapse of the control system. Several typical examples about security threats to modern control systems can be referred to Slammer worm on DavisBesse power plant in Ohio, U.S. (2003) [17], Maroochy Shire Council's sewage control system in Queensland, Australia (2000) [18], and Stuxnet worm which targeted many

industrial control systems [19]. Thus, how to detect and cope with attacks such that an NCS works in a safe environment is an important issue, and has been a hot topic in the past decade [20]–[25].

The five control issues above are fundamental issues for NCSs. A great number of notable results have been reported in the literature. This paper presents a survey of trends and techniques in NCSs, focusing on these five issues, and discusses some challenging issues for future research directions.

Notation: $P \geq 0$ (respectively, $P > 0$), where P is a symmetric matrix, means that P is positive semi-definite (respectively, positive definite). $\text{col}\{\cdot\}$ represents a block-column vector. $\mathbb{N}$ and $\mathbb{R}$ are the sets of positive integers and real numbers, respectively.

II. SAMPLED-DATA CONTROL

In a sampled-data system, a continuous-time physical plant is controlled by a digital controller which generates a sequence of control inputs based on the sampled data packets [26]. In the case that the sampling is periodic, there are fruitful and mature results on analysis and synthesis of sampled-data systems, see, e.g., [27], [28]. However, those results may lose their effectiveness when the sampling is aperiodic. Therefore, during the past two decades, aperiodic sampled-data systems have been an active topic in the field of control, and a number of results have been reported in the literature, see, e.g., survey papers [29], [30] and references therein.

Consider the following linear sampled-data system described by

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ u(t) = Fx(t_k), \quad t \in [t_k, t_{k+1}) \end{cases} \quad (1)$$

where $x(t)$ is the system state and $u(t)$ is the control input; A and B are known real matrices of compatible dimensions while F is the control gain to be designed. An input delay method is well used to analyze stability of the system (1) and design the control gain F . The basic idea is to introduce an artificial piecewise-continuous delay $\tau(t) = t - t_k$. Then the control input $u(t)$ can be rewritten as

$$u(t) = Fx(t - \tau(t)), \quad t \in [t_k, t_{k+1}) \quad (2)$$

which leads to the following time-delay system model of the closed-loop system

$$\dot{x}(t) = Ax(t) + BFx(t - \tau(t)). \quad (3)$$

Suppose that $\max\{t_{k+1} - t_k | k = 0, 1, 2, \dots\} \leq \bar{h}$, where $\bar{h}$ is a known positive constant. Then the input delay $\tau(t)$ satisfies

$$0 \leq \tau(t) \leq \bar{h}, \quad \dot{\tau}(t) = 1, \quad a.e. \quad (4)$$

By choosing a Lyapunov-Krasovskii functional candidate as $V_1(t) = x^T(t)Px(t) + V_R(t)$ with

$$V_R(t) = \int_{-\bar{h}}^0 \int_{t+\theta}^t \dot{x}^T(s)R\dot{x}(s)dsd\theta \quad (5)$$

where $P > 0$ and $R > 0$, some criteria on stability and the existence of suitable control gain F are formulated based on *Lyapunov-Krasovskii stability theorem*. However, those criteria are conservative [31] since they should be satisfied for *any* continuous delay $\tau(t) \in [0, \bar{h}]$, which is a constraint *stricter*

than (4). Inspired by [32], an improved input delay method for sampled-data systems is proposed in [33], where a *time-dependent Lyapunov functional* is constructed as $V_2(t) = x^T(t)Px(t) + V_U(t)$ with

$$V_U(t) = (\hbar - \tau(t)) \int_{t-\tau(t)}^t \dot{x}^T(s)U\dot{x}(s)dsd\theta \quad (6)$$

where $U > 0$. A characteristic of $V_2(t)$ is that it does not grow after the sampling times due to the fact that $V_U(t) > 0$ while it vanishes after the sampling times t_k ($k = 1, 2, \dots$). A stability theorem tailored for such kinds of time-dependent Lyapunov functionals is also established in [33]. The stability theorem also allows one to construct a *discontinuous Lyapunov functional* as $V_3(t) = x^T(t)Px(t) + V_W(t)$ with [34]

$$V_W(t) = \hbar^2 \int_{t_k}^t \dot{x}^T(s)W\dot{x}(s)dsd\theta - \frac{\pi^2}{4} \int_{t_k}^t [x(s) - x(t_k)]^T W [x(s) - x(t_k)] ds \quad (7)$$

where $W > 0$ and $t_k \leq t < t_{k+1}$ ($k = 0, 1, 2, \dots$). The extended Wirtinger inequality ensures that $V_W(t) \geq 0$.

Stability analysis based on the time-independent Lyapunov functional $V_R(t)$, the time-dependent Lyapunov functional $V_U(t)$ and the discontinuous Lyapunov functional $V_W(t)$ can be referred to [34] in detail. It can be found through simulation that a stability criterion based on $V_U(t)$ produces a larger upper bound of $\hbar$ than a stability criterion based on either $V_W(t)$ or $V_R(t)$. A natural question is why the discontinuous Lyapunov functional $V_W(t)$ is introduced. To answer this question, in [34], a sampled-data system is considered by introducing a constant transmission delay into the control input $u(t)$, i.e., $u(t) = Fx(t_k - \eta)$, where $\eta > 0$ is a constant. For such a sampled-data system with a positive constant transmission delay, it is shown that the corresponding time-dependent Lyapunov functional $V_U(t)$ brings *no clear advantages* over the corresponding $V_R(t)$ (see Remark 1 in [34]), while the discontinuous Lyapunov functional $V_W(t)$ is *preferable* since it can deliver less conservative and simpler results than the other Lyapunov functionals $V_R(t)$ and $V_U(t)$. However, from Example 1 in [34], one can see that the obtained results using the discontinuous Lyapunov functional are still conservative. Moreover, it is also illustrated that, by introducing a constant transmission delay into the control input, the upper bound of $\hbar$ is decreased significantly, which reveals the *negative effects* of transmission delays on stability of sampled-data systems.

In 2012, a framework was proposed for stability analysis of sampled-data systems [35]. In this framework, the closed-loop system of the sampled-data system (1) is re-formulated, inspired by the lifting modeling technique. Denote

$$T_k := t_{k+1} - t_k \in [\hbar_m, \hbar_M]$$

where $\hbar_m$ and $\hbar_M$ are two positive constants satisfying $\hbar_m \leq \hbar_M$. Then for $\forall t \in [t_k, t_{k+1})$, one has that $x(t) = \Gamma(t - t_k)x(t_k)$, where

$$\Gamma(t - t_k) := e^{A(t-t_k)} + \int_0^{t-t_k} e^{As} ds BF \quad (8)$$

Let $\sigma := t - t_k \in [0, T_k]$. Define a function $\chi_k(\sigma) := x(t_k + \sigma)$

($k = 1, 2, \dots$). Then $\chi_k(\sigma) = \Gamma(\sigma)\chi_k(0)$, and $\chi_k(\sigma)$ satisfies

$$\begin{cases} \chi_{k+1}(0) = \chi_k(T_k) = \Gamma(T_k)\chi_k(0), & T_k \in [\hbar_m, \hbar_M] \\ \frac{d}{d\sigma}\chi_k(\sigma) = A\chi_k(\sigma) + BF\chi_k(0), & \sigma \in [0, T_k] \end{cases} \quad (9)$$

It is clear that the function $\chi_k(\sigma)$ ($\sigma \in [0, T_k]$) is the state trajectory of the sampled-data system on $[t_k, t_{k+1})$. The first equation in (9) presents the relationship between $x(t_k)$ and $x(t_{k+1})$, which permits the use of the discrete-time stability analysis method. The second equation in (9) reflects the motion of the state trajectory over the period $[t_k, t_{k+1})$. Therefore, the continuous-time method can be employed to investigate the stability of the sampled-data system. The *equivalence* between discrete-time and continuous-time stability analysis methods is disclosed in [35]. The *significance* of the equivalence theorem is that the positive definiteness of the time-dependent Lyapunov functionals and discontinuous Lyapunov functionals can be relaxed by a *looping condition*. According to this equivalence theorem, a Lyapunov functional candidate can be constructed as $V_4(t) = x^T(t)Px(t) + \mathcal{V}_0(\sigma, \chi_k)$, where $\mathcal{V}_0 : [0, \hbar_M] \times \mathbb{K} \rightarrow \mathbb{R}$ is a continuous and differentiable functional satisfying

$$\mathcal{V}_0(T, z(\cdot)) = \mathcal{V}_0(0, z(\cdot)), \quad \forall T \in [\hbar_m, \hbar_M], \quad \forall z \in \mathbb{K}, \quad (10)$$

which is a looping condition. In this sense, $\mathcal{V}_0(\sigma, \chi_k)$ is called a *looped functional*. A looped functional suggested in [35] is given as

$$\begin{aligned} \mathcal{V}_0(\sigma, \chi_k) = & (T_k - \sigma) \left\{ \xi_k^T(\sigma) [S_1 \xi_k(\sigma) + 2S_2 \chi_k(0)] \right. \\ & \left. + \int_0^\sigma \dot{\chi}_k^T(s) R \dot{\chi}_k(s) ds + \sigma \chi_k^T(0) X \chi_k(0) \right\} \end{aligned} \quad (11)$$

where $\xi_k(\sigma) = \chi_k(\sigma) - \chi_k(0)$. Since $\mathcal{V}_0(0, \chi_k) = \mathcal{V}_0(T_k, \chi_k)$ holds for $\forall T_k \in [\hbar_m, \hbar_M]$, the functional $\mathcal{V}_0(\sigma, \chi_k)$ satisfies the looping condition (10) while the real matrices S_1, S_2, X and R are *not* necessarily positive definite. Revert $\mathcal{V}_0(\sigma, \chi_k)$ to a function on time t to obtain

$$\begin{aligned} \tilde{\mathcal{V}}(t) = & \mathcal{V}_0(\sigma, \chi_k) \\ = & (t_{k+1} - t) \begin{bmatrix} \int_{t_k}^t \dot{x}(s) ds \\ x(t_k) \end{bmatrix}^T \begin{bmatrix} S_1 & S_2 \\ S_2^T & (t - t_k)X \end{bmatrix} \begin{bmatrix} \int_{t_k}^t \dot{x}(s) ds \\ x(t_k) \end{bmatrix} \\ & + (t_{k+1} - t) \int_{t_k}^t \dot{x}^T(s) R \dot{x}(s) ds, \quad t \in [t_k, t_{k+1}). \end{aligned} \quad (12)$$

By taking such a looped functional, some less conservative stability conditions are derived in [35]. Borrowing the above idea, a *two-sided* looped functional is introduced in [36] in the sense that information on both states at two vertices of the interval $[t_k, t_{k+1})$, i.e. $x(t_k)$ and $x(t_{k+1})$, is exploited. It reads as $V_5(t) = x^T(t)Px(t) + \mathcal{V}(t)$, where

$$\begin{aligned} \mathcal{V}(t) = & \begin{bmatrix} \eta_1(t) \\ \eta_2 \end{bmatrix}^T \begin{bmatrix} DQ_1 & DQ_2 + X \\ Q_2^T D + X^T & (t_{k+1} - t)(t - t_k)Z \end{bmatrix} \begin{bmatrix} \eta_1(t) \\ \eta_2 \end{bmatrix} \\ & + (t_{k+1} - t) \int_{t_k}^t \dot{x}^T(s) R_1 \dot{x}(s) ds \\ & - (t - t_k) \int_t^{t_{k+1}} \dot{x}^T(s) R_2 \dot{x}(s) ds \end{aligned} \quad (13)$$

with $\mathcal{D} := \text{diag}\{(t_{k+1} - t)I, (t - t_k)I\}$ and

$$\eta_1(t) := \begin{bmatrix} \int_{t_k}^t \dot{x}(s) ds \\ \int_t^{t_{k+1}} \dot{x}(s) ds \end{bmatrix}, \eta_2 := \begin{bmatrix} x(t_k) \\ x(t_{k+1}) \end{bmatrix}.$$

Compared with $\tilde{\mathcal{V}}(t)$, the two-sided looped functional $\mathcal{V}(t)$ includes such an integral from t to t_{k+1} . Moreover, the state $x(t_{k+1})$ is also included in $\mathcal{V}(t)$. By employing the second-order Bessel-Legendre inequality [37] to bound the integral terms $-\int_{t_k}^t \dot{x}^T(s) R_1 \dot{x}(s) ds$ and $-\int_t^{t_{k+1}} \dot{x}^T(s) R_2 \dot{x}(s) ds$ appearing in the derivative of $\mathcal{V}(t)$, the obtained stability criteria are of less conservatism than some existing ones as those in [35]. Nevertheless, it is not an easy task to design suitable control gains F due to the fact that the system matrices are coupled with a number of matrix variables in the stability criteria [36].

III. QUANTIZATION CONTROL

Signals should be coded via analog-to-digital coders before the signals are released to a communication network. On the one hand, an analog-to-digital coder is usually employed to convert a continuous-time analog signal to its corresponding discrete-time digital one [38]. The error between the actual analog value and the converted digital one is usually unavoidable due mainly to the operation of rounding or truncation. On the other hand, data compression of sampled signals is quite common in order to reduce network loads such that limited channel bandwidth can be utilized effectively. As typical coders, quantizers have gained ever-increasing interest of researchers. Essentially, a quantizer is a symmetrical piecewise function with respect to the origin, mapping input values to output values with an infinite countable number of levels [39]. Generally, there are *uniform quantizers* and *logarithmic quantizers*.

Denote a quantizer as $f(\cdot)$ and the number of quantization levels as $\#g(\varepsilon)$ over the interval $[\varepsilon, 1/\varepsilon]$. Its density is defined as [39], [40]

$$\eta_f = \limsup_{\varepsilon \rightarrow 0} \frac{\#g(\varepsilon)}{-\ln \varepsilon}.$$

With this definition, a logarithmical increase of the number of quantization levels occurs if the length of interval $[\varepsilon, 1/\varepsilon]$ gets larger. A small value of η_f corresponds to a coarse quantizer. Especially, η_f becomes zero if the number of quantization levels is finite, while η_f equals infinity if the quantizer is linear. A classical approach to investigating quantization effects is to treat the quantization errors as unknown but bounded disturbance or norm bounded uncertainties. In doing so, classical robustness analysis tools or bounded stability theory can be employed to address various quantization-induced issues. Furthermore, the desired system performance can be guaranteed by adjusting the error bound or the quantization density, and the desired controller or filter parameters can be obtained simultaneously.

A. Uniform Quantizer

Suppose that the overall quantizer range is $[-L, L]$ with $L > 0$. Denote by M the quantizer level length. Let $M = 2L/(2^b - 1)$, where b is the number of bits for digital sensors. Then the uniform quantizer is given by [41]

$$f(y) = y + q_k \quad (14)$$

where $f(y)$ is the quantized signal and q_k is the quantization error process. The quantization error process q_k is usually modeled as an unknown but bounded disturbance or an additive white uniform distributed noise with each element being uniformly distributed over $[-0.5M, 0.5M]$. For a stochastic model, the variance of such a quantization error process is $(M^2/12)I$.

The quantization effect is investigated in [42] with the model above for systems with time-varying delays and input saturation. Both the domain of attraction and the reachable ellipsoid are derived by using a Lyapunov-Krasovskii functional method. It should be pointed out that the obtained reachable ellipsoid is dependent on the quantization error bound $0.5M$. Recently, some quantized control issues are addressed by using different control approaches. For example, networked nonlinear systems subject to the try-once-discard protocol scheduling are analyzed in the framework of exponentially ultimate boundedness [43]. Uncertain nonlinear systems subject to input constraints are investigated in the framework of approximate dynamic programming [44]. Moreover, a robust analysis method is developed in the case of imprecise quantizer level length [45]. For consensus control of multi-agent systems, there are a number of preliminary results available in the literature, see, e.g., [46], [47] and references therein.

Dynamic quantizers can be employed to guarantee desired closed-loop system performance. A dynamic quantizer is based on a dynamic quantizer function $f_\mu(y) = \mu f(y/\mu)$, where μ is a dynamic variable referred to as a “zoom” variable, and f is a certain quantizer. The zoom variable μ is used to adjust both the quantizer range and the quantizer error bound based on the value of output measurement y . The simplified form of a dynamic quantizer is described by [48]–[50]

$$\|f_\mu(y) - y\| \leq \delta\mu, \text{ if } \|y\| < L\mu \quad (15)$$

where $\delta\mu$ is the maximum quantization error for all y in the range of the quantizer. Based on the dynamic quantizer above, event-triggered input-to-state stability with a size-adjustable set is investigated in [50]. Moreover, by introducing an adjustable “center” and some “zoom” parameters, a coarse quantizer is studied in [51] to achieve the input-to-state stability with respect to external disturbance. However, when communication protocols are involved, investigating the quantization effects on NCSs becomes more complex and deserves further effort [52]–[54].

B. Logarithmic Quantizer

For the predetermined set of quantization levels

$$\mathcal{U} = \{\pm u_i | u_i = \rho^j u_0, j = 0, \pm 1, \pm 2, \dots\} \cup \{0\}$$

with $0 < \rho \leq 1$ and $u_0 > 0$, a logarithmic quantizer $f(\cdot)$ is described by [40], [55]

$$f(y) = \begin{cases} u_i, & \frac{u_i}{1+\kappa} < y \leq \frac{u_i}{1-\kappa}, \\ 0, & y = 0, \\ -f(-y) & y < 0 \end{cases} \quad (16)$$

with $\kappa = (1 - \rho)/(1 + \rho)$. For such a logarithmic quantizer, it is not difficult to verify that its quantization density satisfies $\eta_f = 2/\ln(1/\rho)$. In this sense, the quantized measurement is modeled as

$$f(y) = (1 + \Delta(k))y \quad (17)$$

where $\Delta(k) \in [-\kappa, \kappa]$.

There are three kinds of methods to analyze the system performance under a logarithmic quantizer. The first method is based on a quantization-dependent Lyapunov function [55], [56]. Using this method, the obtained conditions depend closely on the parameter κ , leading to some less conservative results on control performance. The second one is called a robust analysis method. If one defines $F(k) = \kappa^{-1}\Delta(k)$, then $F(k)$ is an unknown real-valued time-varying matrix $F(k)$ satisfying $F^T(k)F(k) \leq I$. Thus, the quantization control problem can be transformed into the well-known robust control/filtering problem [57], [58]. Based on the robust analysis method, a scheme with the form of coupled recursive Riccati difference equations is developed in [58] to deal with the H_∞ control of time-varying systems over finite-horizon. It is worth mentioning that this method is similar to the direct utilization of the constraint $\|\Delta(k)\| \leq \kappa$, as is done in [59] for recursive filtering. The third method is related to the sector representation of the quantization error of a logarithmic quantizer. The quantization error $q_k(y)$ can also be characterized by the following sector condition

$$q_k^T(y)(q_k(y) - 2\kappa y) \leq 0. \quad (18)$$

Then the quantized measurement is rewritten as

$$f(y) = y + q_k(y). \quad (19)$$

Some well-known control methods using the above sector condition can be employed to cope with the problem of quantization control, see, e.g., [60] and [61] in detail.

The analysis above just takes a snapshot for quantization control, including two quantizer models and research methods. Some other complex quantization models are not involved in this paper. For example, a logarithmic-hysteretic quantizer is proposed in [62] to describe both the quantization and the deadzone. A logarithmic-uniform quantizer and a hysteresis-uniform quantizer are designed in [63], where a stability analysis approach is developed by constructing a compensation scheme for the effects of the state quantization, and the system performance can be improved by appropriately adjusting design parameters.

IV. NETWORKED CONTROL

In an NCS, sensor data and control signals are transmitted via communication channels from Sensor to Controller (S-C) and from Controller to Actuator (C-A). Due to the limitation of network bandwidth, there exist inevitable network traffic and congestion, thus causing some network-induced phenomena during data transmission. Network-induced delays and packet dropouts are two common phenomena that an NCS encounters, and they are usually regarded as a factor that may degrade system performance or jeopardize the system stability. Thus, how to deal with network-induced delays and

packet dropouts has been a fundamental issue, and a great number of results on this topic have been published in the literature, see some monographs [64], [65] and survey papers [4], [7]–[9], [66], [67].

In this section, we present a review of two common models of network-induced delays and packet dropouts, namely *random model* and *input delay model*, depending on the random and time-varying characteristics of network-induced delays and packet dropouts.

A. Random Model

A random model of network-induced delays and packet dropouts is quite often and natural. This model is usually based on the discrete-time description of the physical plant. Suppose that the S-C and C-A channels have different random attributions. A random model for network-induced delays is established in [68]. Let $\tau_k \in [0, \tau]$ and $d_k \in [0, d]$ be the random time delays in the S-C channel and the C-A channel, respectively, where τ and d are two known positive integers. A reasonable assumption is made in [68] that two random sequences $\{\tau_k\}$ and $\{d_k\}$ are homogeneous Markov chains that take values in $\{0, 1, \dots, \tau\}$ and $\{0, 1, \dots, d\}$ with transition probability matrices $[\lambda_{ij}]$ and $[\pi_{rs}]$, respectively. Under such an assumption, the controller is of the form

$$u(k) = F(\tau_k, d_{k-1})x(k - \tau_k - d_k). \quad (20)$$

It is claimed that the model above of network-induced delays takes packet dropouts into account if $\lambda_{ij} = 0$ or $\pi_{rs} = 0$ occurs for $j > i + 1$ or $s > r + 1$. The advantage of the model above is that the control gain F is not a constant real matrix but a variable one dependent on network-induced delays τ_k and d_{k-1} .

In [69], [70], random delays in both S-C and C-A channels are modeled as a linear function of a stochastic variable satisfying Bernoulli random binary distribution. Suppose that $\delta \in \mathbb{R}$ is a stochastic variable being a Bernoulli distributed white sequence with $\text{Prob}\{\delta=1\}=\bar{\delta}$ and $\text{Prob}\{\delta=0\}=1-\bar{\delta}$. The following description of sensor measurement $y_{c,k}$

$$\begin{cases} y_k = Cx_k \\ y_{c,k} = (1-\delta)y_k + \delta y_{k-1} \end{cases} \quad (21)$$

provides a random model of network-induced delays, where x_k and y_k are the system state and the system output, respectively; and C is a constant real matrix. The output y_k is sent to an observer subject to the network-induced delay τ_d . If τ_d is less than one sampling period, $y_{c,k} = y_k$. If τ_d is longer than one sampling period but less than two sampling periods, $y_{c,k} = y_{k-1}$. If δ takes the value of 1 during more than two consecutive sampling periods, it is true that τ_d is longer than two sampling periods.

The idea above is used to model random packet dropouts [71]. In the S-C channel, let the stochastic variable $\alpha_k \in \mathbb{R}$ be a Bernoulli distributed white sequence. Then the measurement output y_k is described by

$$y_k = \alpha_k Cx_k + Dw_k \quad (22)$$

where w_k is the disturbance input, and C and D are constant real matrices. $\alpha_k = 0$ means that a packet dropout occurs. This measurement output y_k is sent to an observer-based controller

to generate control signal $\hat{u}_k$. Then the control signal $\hat{u}_k$ is transmitted to actuator via the C-A channel. The random packet dropout phenomenon in the C-A channel is modeled as a stochastic variable $\beta_k \in \mathbb{R}$, which is mutually independent of α_k . The stochastic variable β_k is also a Bernoulli distributed white sequence. Then the control input u_k to actuate the physical plant is given as $u_k = \beta_k \hat{u}_k$. If $\beta_k = 0$, then $u_k = 0$, which means that the packet is lost during transmission.

Note that the model (21) is available just for random network-induced delays while the model (22) just for random packet dropouts. A unified model to describe both random network-induced delays and packet dropouts is presented in [72]. The idea is to introduce a number of indicator functions $\mathcal{I}_{\{s_k=0\}}$ and $\mathcal{I}_{\{s_k=d_j\}}$ ($j = 1, 2, \dots, q$), where s_k is a stochastic variable to determine how large the network-induced delay and the possibility of data missing could be at the time instant k . The stochastic variable s_k is no longer a Bernoulli distributed white sequence, while its mathematical expectation satisfies $\mathbb{E}\{\mathcal{I}_{\{s_k=0\}}\} = \text{Prob}\{s_k=0\} = p_0$ and $\mathbb{E}\{\mathcal{I}_{\{s_k=d_j\}}\} = \text{Prob}\{s_k=d_j\} = p_j$ with $\sum_{j=0}^q p_j \leq 1$, where p_j ($j = 0, 1, \dots, q$) are known non-negative constants. In this situation, the system measurement is given by

$$y_k = \mathcal{I}_{\{s_k=0\}} C_0 x_k + \sum_{j=1}^q \mathcal{I}_{\{s_k=d_j\}} C_j x_{k-d_j} + D w_k. \quad (23)$$

Based on this model, if $\sum_{j=0}^q p_j < 1$, the probability of packet dropout is $1 - \sum_{j=0}^q p_j$; otherwise, no packet dropout occurs.

The analysis above provides several typical stochastic methods to model random network-induced delays and packet dropouts. Although there are different approaches reported in the literature to dealing with the same problem, the basic ideas of them are similar to the ones mentioned above. In the recent years, these methods have stirred up much interest of researchers in the field and have been widely used to handle a number of issues, such as input-to-state stability, input-output stability, H_∞ control, Kalman filtering, H_∞ filtering, fault-tolerant control and so on.

B. Input Delay Model

An input delay model is well-used to deal with network-induced delays and packet dropouts. Suppose that the controller and actuator are event-driven while the sensor is time-driven with a period $T_s > 0$. Each sampled signal from the sensor is stamped by time, and encapsulated together with its time-stamp into a packet, say $(j, x(jT_s))$ ($j = 1, 2, \dots$). The packet $(j, x(jT_s))$ is transmitted to the controller via the S-C channel and ultimately to the actuator via the C-A channel. If the transmission is successful, the time-consuming denoted by τ_j from the sensor to the actuator is called a network-induced delay, which is lumped together from both channels. Thus, the time-instant when the packet arrives at the actuator can be calculated as $jT_s + \tau_j$. At this time, if state feedback is concerned, the control input that actuates the physical plant can be given by

$$u(t)|_{t \rightarrow (jT_s + \tau_j)^+} = Kx(jT_s). \quad (24)$$

This control input will be changed only when the next new packet arrives at the actuator. Since τ_j is not necessarily equal

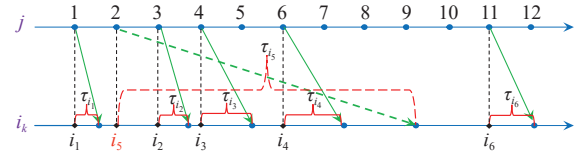


Fig. 1. The diagram of time scheduling.

to τ_q for $q \neq j$, the next new packet transmitted successfully may be not $(j+1, x((j+1)T_s))$. Denote by $\{i_1, i_2, i_3, \dots\}$ the time-stamp sequence indicating that those sampled packets arrive at the actuator *in order*. Then it is possible that $i_k > i_{k+1}$, but it must be true that $i_k T_s + \tau_{i_k} < i_{k+1} T_s + \tau_{i_{k+1}}$. Fig. 1 is an example to show the time-stamp sequence $\{i_1, i_2, i_3, \dots\}$. That is, $i_1 = 1, i_2 = 3, i_3 = 4, i_4 = 6, i_5 = 2$ and $i_6 = 7$. Clearly, $i_4 > i_5$ while $i_4 T_s + \tau_{i_4} < i_5 T_s + \tau_{i_5}$. Therefore, the control input can be described by

$$u(t) = Kx(i_k T_s), \quad i_k T_s + \tau_{i_k} \leq t < i_{k+1} T_s + \tau_{i_{k+1}}. \quad (25)$$

Define a piecewise continuous function

$$\tau(t) = t - i_k T_s, \quad t \in [i_k T_s + \tau_{i_k}, i_{k+1} T_s + \tau_{i_{k+1}}). \quad (26)$$

Then the control input $u(t)$ can be rewritten as

$$u(t) = Kx(t - \tau(t)), \quad i_k T_s + \tau_{i_k} \leq t < i_{k+1} T_s + \tau_{i_{k+1}}. \quad (27)$$

It is clear that $\tau(t)$ includes information on network-induced delays. Moreover, information on packet dropouts is also included in $\tau(t)$. In fact, if $\{i_1, i_2, i_3, \dots\} \neq \mathbb{N}$, then those packets whose time-stamps belong to the complement set of $\{i_1, i_2, i_3, \dots\}$ are dropped out; otherwise, no packet dropouts happen. This model $\tau(t)$ has been widely used in NCSs, which is referred to as an *input delay method* [73], [74].

In the model above, the packet disorder phenomenon is non-negligible. A packet disorder occurs if the packet sent earlier arrives at the actuator later. That is, if $i_{k+1} < i_k$, the packet with stamp i_{k+1} is transmitted *in disorder*, see for example the packet $(i_5, x(i_5 T_s))$ in Fig. 1. As a result, the network-induced delay is very big (e.g. τ_{i_5} is much bigger than others). Let us consider $\tau(t)$ on the time interval $t \in [i_5 T_s + \tau_{i_5}, i_6 T_s + \tau_{i_6})$. It is clear that $\tau_{i_5} \leq \tau(t) < (i_6 - i_5) T_s + \tau_{i_6}$, which leads to large lower and upper bounds of $\tau(t)$, definitely making the obtained stability conditions *infeasible*. Keeping this in mind, a *logic* zero-order-hold mechanism is introduced to *actively* drop out those disordered packets [75]. It can be done if the ZOH *chooses* the latest packet to store, see [75] in detail. The introduction of a logic ZOH provides a feasible way for active dropouts. Such an active dropout can ensure $i_{k+1} > i_k$, which makes the input delay method more useful in the stability analysis and synthesis of NCSs.

Another model can be referred to [76]. The basic idea behind this model is based on the assumption that the properties of both two delays from the S-C channel and the C-A channel may not be identical. Denote by $d_s(t)$ and $d_a(t)$, respectively, the time-varying delays induced from the S-C channel and the C-A channel. Two network-induced delays $d_s(t)$ and $d_a(t)$ may *not* be combined together since they are of different properties (for instance, one is differentiable while one is non-differentiable). In this situation, the control input (for state feedback) can be described as

$$u(t) = Kx(t - d_s(t) - d_a(t)). \quad (28)$$

Then the closed-loop NCS is modeled as a system with *two additive time-varying delay components*. Although plenty of results on NCSs have been obtained based on this model, how these two delays reflect some exact features of communication networks has not been discussed yet.

The input delay model above has an *implicit* assumption that there are only one sensor, one controller and one activator in the NCS with no competition of them when gaining access to networks. If there are more network nodes and only one node per transmission is allowed to transmit its packet at any time, proper communication scheduling (or communication protocol) among the nodes is necessary. Arguably, there are two communication protocols available: static and dynamic. A static communication protocol, like Round-Robin (RR), periodically assigns one node in a *prefixed order* to transmit its packet, while a dynamic communication protocol, like Try-Once-Discard (TOD), dynamically assigns one node to transmit its packet based on the *network-induced errors* [77]. Under either an RR protocol or a TOD protocol, it is still challenging to establish such an input delay model as above to handle network-induced delays and packet dropouts simultaneously. For this reason, in [77], it is assumed that no network-induced delays and packet dropouts occur, and in [78], [79], network-induced delays are allowed while packet dropouts are not expected when RR protocols or TOD protocols are involved.

The random and input delay models aforementioned are commonly used to describe both network-induced delays and packet dropouts. Based on these two models, the related closed-loop NCS is modeled as a number of systems either in the continuous-time domain or in the discrete-time domain, such as stochastic systems, time-delay systems, switched systems, Markov jumping systems, impulsive systems, and hybrid systems, and thus leading to a large number of results on NCSs in the open literature. Although those results may reveal more or less effects of network imperfections on NCSs, there is much room for improvement. How to build an exact mathematical model for network imperfections, especially for packet dropouts, such that the effects of them on NCSs can be disclosed explicitly is of much significance and importance both in theory and in practice.

V. EVENT-TRIGGERED CONTROL

Event-triggered control has been long in existence, which can date back to 1960 for the research of an accelerometer with pulse feedback [80], where *pulse generators* are used to move a pendulum towards the center position once a deviation is detected. Such pulse generators embrace the basic idea of event-triggered control. In 1993, it was pointed out that ‘*control is not executed unless it is required*’ when answering the question ‘*Should responsive systems be event-triggered or time-triggered?*’ [81]. Unfortunately, this idea did not attract much research interest at that time. Seemingly, it is pretty simple and natural, but its theoretical analysis is quite complicated. An encouraging result contributes to the experimental analysis on a first-order stochastic system, which shows that event-triggered control has a number of advantages

compared to time-triggered control [82]. However, the study on event-triggered control is still sporadic until a general solution to the problem of stability is derived for event-triggered control systems [14]. Boosted by the results, event-triggered control has been a quite hot topic in the field of NCSs even up until now, see, e.g., survey papers [83]–[85] and references therein.

Event-triggered control is a natural way in the sense that a control execution depends on an event rather than the elapse of a fixed time period. An event-generator is used to detect if an event is triggered before control executes. In an NCS, the event-generator is usually embedded into the sensor to decide whether or not the current signal should be sampled, leading to *event-triggered sampling* (Fig. 2(a)). Alternatively, an event-generator is located behind a sensor to determine if sampled signals should be released to the communication network to transmit, resulting in *event-triggered transmission* (Fig. 2(b)). Event-triggered sampling is an ideal approach to choosing only necessarily sampled signals to carry out some desired control tasks such that communication resources, energy resources for battery-based devices, and network resources can be saved to a maximum extent. Nevertheless, event-triggered sampling is liable to fall into Zeno behaviour, which means that infinite sampling times occur within a certain finite time-interval, especially when the system is disturbed by external noises. Event-triggered transmission, however, is based on either periodically or aperiodically sampled data, and thus the Zeno behaviour is not involved. In this sense, such an event-triggered transmission scheme is also known as a sampled-data-based event-triggered transmission scheme [86]–[91], which aims at reducing network loads to save precious network communication resources. It is noteworthy that the sampled-data-based event-triggered transmission scheme can save precious network resources while computation burdens and energy consuming of sensors may not be reduced [30]. In the recent years, event-triggered sampling and event-triggered transmission both have received much attention. The key point is how to devise a suitable condition to generate an event. In this section, we provide a brief summary of recent developments on devising event-triggering schemes (ETSs) for both event-triggered sampling and event-triggered transmission.

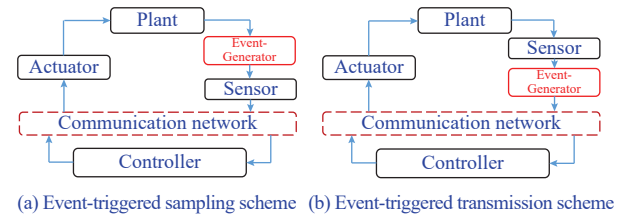


Fig. 2. Two schemes of event-triggered control.

A. ETSs for Event-triggered Sampling

An ETS is originally motivated by Lyapunov stability. Consider the control system described by

$$\dot{x}(t) = f(x(t), u(t)) \quad (29)$$

where $x(t)$ and $u(t)$, respectively, are the system state and the

control input, respectively. A suitable controller $u(t) = \mu(x(t))$ is designed *a priori* such that the resultant closed-loop system

$$\dot{x}(t) = f(x(t), \mu(x(t) + e(t))) \quad (30)$$

is input-to-state stable (ISS) with respect to e , where e is the error vector. Then there exists an ISS Lyapunov function $V(x)$ satisfying

$$\rho_1(\|x\|) \leq V(x) \leq \rho_2(\|x\|) \quad (31)$$

$$\dot{V}(x)|_{(30)} \leq -\alpha(\|x\|) + \gamma(\|e\|) \quad (32)$$

where $\rho_1(\cdot), \rho_2(\cdot), \alpha(\cdot)$ and $\gamma(\cdot)$ are class $\mathcal{K}_\infty$ functions¹.

Event-triggered sampling aims to provide a scheme to sample the system state such that $\dot{V}(x) < -\bar{\alpha}(\|x\|) \leq 0$ for all $t \geq 0$, where $\bar{\alpha}(\cdot)$ is a class $\mathcal{K}_\infty$ function. We denote the sampling sequence by

$$0 = t_0, t_1, t_2, \dots, t_k, t_{k+1}, \dots \quad (33)$$

Then $x(t) = x(t_k)$ for $t_k \leq t < t_{k+1}$. Set $e(t) = x(t) - x(t_k)$, $t \in [t_k, t_{k+1})$. Suppose that the communication network has ideal quality of service. Then the control input can be given by

$$u(t) = u(t_k) = \mu(x(t_k)) = \mu(x(t) + e(t)), t \in [t_k, t_{k+1}) \quad (34)$$

In order to maintain the system stability, an event-triggering condition can be devised as

$$\sigma\alpha(\|x(t)\|) \leq \gamma(\|e(t)\|) \quad (35)$$

where the threshold $\sigma \in (0, 1)$ is a constant. Once the condition (35) is satisfied, an event, i.e., sampling the system state, is immediately invoked. In this sense, the sampling instant can be determined by

$$\begin{cases} t_0 = 0 \\ t_{k+1} = \inf\{t > t_k | \sigma\alpha(\|x(t)\|) \leq \gamma(\|e(t)\|)\} \end{cases} \quad (36)$$

Such an ETS can deduce the following inequality

$$\dot{V}(x)|_{(30)} \leq -\alpha(\|x\|) + \gamma(\|e\|) \leq -(1 - \sigma)\alpha(\|x(t)\|) \quad (37)$$

which ensures the stability of the closed-loop system under the sampling scheme due to $\sigma \in (0, 1)$.

The event-triggering condition (35) depends on the system state $x(t)$ and its error $e(t)$ rather than a fixed time period. Thus, signals are sampled only when necessary. In what follows, we give a detailed explanation about the condition.

First, note that $\alpha(\cdot)$ and $\gamma(\cdot)$ are continuous and strictly increasing functions, and that $e(t_k) = 0$. Hence, at t_k , one has $\sigma\alpha(\|x(t_k)\|) > \gamma(\|e(t_k)\|) = 0$. According to continuity, when t is increasing from t_k , the inequality $\sigma\alpha(\|x(t)\|) > \gamma(\|e(t)\|)$ holds. Second, if the closed-loop system is deviating from its stability, $x(t)$ will be getting larger, leading to the increase of both $\sigma\alpha(\|x(t)\|)$ and $\gamma(\|e(t)\|)$. Once $\sigma\alpha(\|x(t)\|) = \gamma(\|e(t)\|)$ is satisfied, a new event time t_{k+1} is generated such that $e(t)$ returns to zero, i.e. $e(t)|_{t=t_{k+1}} = 0$, which turns the closed-loop system back to stability, see Fig. 3.

The ETS above is known as a *static* ETS. To relax the event-triggering condition (35), a *dynamic* ETS is proposed in [92]. The basic idea is to introduce an internal dynamic variable $\eta(t)$ in (35) such that

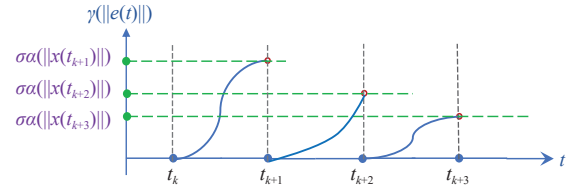


Fig. 3. Two schemes of event-triggered control.

$$\eta(t) + \theta[\sigma\alpha(\|x(t)\|) - \gamma(\|e(t)\|)] \leq 0 \quad (38)$$

where $\theta > 0$ is a parameter to be designed, and $\eta(t)$ satisfies

$$\dot{\eta} = -\beta(\eta) + \sigma\alpha(\|x\|) - \gamma(\|e\|), \eta(0) = \eta_0 \quad (39)$$

where $\beta(\cdot)$ is a class $\mathcal{K}_\infty$ function and $\eta_0 > 0$.

Under the dynamic ETS (38), it is proven that $\eta(t) \geq 0$ for $t \geq 0$. At t_k , $\sigma\alpha(\|x(t_k)\|) > \gamma(\|e(t_k)\|) = 0$. Let t increase slowly from t_k . Then $\sigma\alpha(\|x(t)\|) > \gamma(\|e(t)\|)$. Once $\sigma\alpha(\|x(t)\|) = \gamma(\|e(t)\|)$ is satisfied, the dynamic ETS (38) allows t to keep going until (38) holds, which means that the inequality $\sigma\alpha(\|x(t)\|) \geq \gamma(\|e(t)\|)$ does not need to be always satisfied for $t \geq 0$. In this sense, the minimum inter-event time $\min\{t_{k+1} - t_k | k = 0, 1, 2, \dots\}$ is larger than that of the static ETS (35). On the other hand, if rewriting (38) as

$$\sigma\alpha(\|x(t)\|) - \gamma(\|e(t)\|) \leq -\frac{\eta(t)}{\theta}$$

the dynamic ETS reduces to the static ETS (35) if $\theta \rightarrow \infty$. Such a dynamic ETS is refined in [93] to deal with the distributed set-membership estimation issue for a class of discrete-time linear time-varying systems over a wireless multi-sensor network. It is analytically and numerically shown that such a dynamic ETS can lead to larger average inter-event time and thus less totally released data packets.

Note that both the static ETS (36) and the dynamic ETS (38) cannot guarantee a positive minimum inter-event time to avoid Zeno behaviour, especially for perturbed physical plants [15]. In this situation, a mixed time- and event-triggered scheme is proposed in [94], where the sampling instant is calculated by

$$\begin{cases} t_0 = 0 \\ t_{k+1} = \inf\{t > t_k + T | \mathcal{F}(\sigma, x, e) \leq 0\} \end{cases} \quad (40)$$

with $T > 0$ denoting a constant and $\mathcal{F}(\sigma, x, e)$ representing a certain function with respect to σ, x and e . Clearly, with this scheme, a positive minimum inter-event time greater than T is guaranteed even though external disturbance is imposed on the system. This scheme is also studied in [95] by presenting a switching model for the closed-loop system. Mixed time- and event-triggered schemes have strong application background. In fact, in non-rush hours, such a mixed triggering scheme can be used to control the traffic lights' change. Once a car arrives just before a stopping line, an event should be triggered to turn the light from red to green. Such a task can be completed within a fixed time T after the event is triggered.

A common feature of the ETSs above is that the system state $x(t)$ should be monitored at any time. To carry out such a task, extra hardware should be embedded into the system such that an ETS works well. This definitely incurs an increase of cost. In this situation, self-triggered control comes to the fore

¹A function $\alpha(\cdot)$ is said to be of class $\mathcal{K}_\infty$ if it is continuous, strictly increasing and satisfies $\lim_{s \rightarrow \infty} \alpha(s) = \infty$ and $\alpha(0) = 0$.

[96]. The concept of self-trigger means that the next sampling time can be *predicted* by the current sampled signal, which can be generally described by

$$t_{k+1} = t_k + \mathcal{F}(x(t_k)) \quad (41)$$

where $\mathcal{F}(x(t_k))$ is a prediction function with respect to $x(t_k)$. It is clear that a self triggering scheme allows us to compute the next time t_{k+1} *offline* based on the available state $x(t_k)$. However, how to define an exact prediction function is a challenging issue. Based on an ISS Lyapunov function and $\mathcal{L}_2$ gain performance, a prediction function is defined as [96]

$$\mathcal{F}(x(t_k)) = \frac{1}{\alpha} \ln \left(1 + \delta \alpha \frac{\rho(x(t_k))}{\mu_0(x(t_k))} \right) \quad (42)$$

where $\alpha, \delta, \rho(\cdot)$ and $\mu_0(\cdot)$ are referred to [96]. For exponential ISS, a different prediction function is designed in [97].

ETSs for event-triggered sampling have gained considerable attention during the past decade. A large number of results have been reported on ETSs for linear or nonlinear or stochastic systems in either continuous-time or discrete-time domain, see some recent results, e.g., [98]–[106]. However, most results are derived from the digital implementation perspective. First, design a suitable controller such that the closed-loop system has some proper performance. Then, design an ETS to implement the predesigned controller in a digital platform such that the closed-loop performance can be retained while the number of sampling can be reduced as much as possible. This represents a kind of *passive* event-triggered control schemes since the ETS should cater for the predesigned controller. *Active* event-triggered control is to actively design a suitable controller for the physical plant under a certain ETS. That is, the controller is designed to cater for the ETS. As such, some closed-loop performance can be ensured while the number of sampling and/or transmission can be reduced to the minimum extent. Nevertheless, such an *active* event-triggered control scheme remains challenging although is of great practical significance. Another point is that event-triggered control should take communication constraints like network-induced delays and packet dropouts into account, while just few results on them are reported, see, e.g., [99], [100].

B. ETSs for Event-triggered Transmission

An ETS for event-triggered transmission aims at saving precious network resources by reducing the number of data packets before being released to a communication network. It works at every sampling instant to choose necessary data packets to transmit through networks. Suppose that the sequence of sampling instants is $\{ih | i = 1, 2, \dots\}$, where $h > 0$ is the sampling period. The event-triggering condition for an ETS is usually devised as follows [16]

$$\begin{aligned} \psi^T(t_k, j) \Omega \psi(t_k, j) &\geq \varepsilon x^T(t_k h) \Omega x(t_k h) \\ \psi(t_k, j) &= x((t_k + j)h) - x(t_k h) \end{aligned} \quad (43)$$

where $\varepsilon \in (0, 1)$ is a threshold and Ω a weighting symmetric positive definite matrix; and $t_k h$ is the latest time instant when a data-packet is triggered. Based on this condition, the next time instant $t_{k+1} h$ can be determined by

$$\begin{cases} t_0 = 0 \\ t_{k+1} h = t_k h + \min\{jh | (43) \text{ holds}, j = 1, 2, \dots\}. \end{cases} \quad (44)$$

Analysis in depth on the ETS (44) can be found in [84]. As a complementary, we present one more insight to the condition (43) compared with (35) and (38). A clear observation on the event-triggering conditions (35) and (38) is that they are closely dependent on the derivative of a certain Lyapunov function. This is because they should be designed to passively cater for some preset closed-loop performance. However, the condition (43) is independent of any Lyapunov function and its derivative. The objective of (43) is to reduce the number of sampled data packets. According to the real-time status of network traffic, one can adjust the threshold to release more or less data packets to the network. Then, based on the ETS, we actively design suitable controllers such that the closed-loop system can achieve some expected performance. In this sense, the condition (43) is more practical than those (35) and (38). Thus, it has been widely used to deal with a number of control issues over networks, such as H_∞ control, $\mathcal{L}_2$ control, sliding mode control, fuzzy control, adaptive control, stochastic control, distributed control, fault diagnosis, state estimation and filtering, consensus and formation control.

Recently, the event-triggering condition (43) is modified as

$$\psi^T(t_k, j) \Omega_1 \psi(t_k, j) \geq \varepsilon x^T(t_k h) \Omega_2 x(t_k h) \quad (45)$$

where Ω_1 and Ω_2 are two different weighting symmetric positive definite matrices [107], [108]. The introduction of the weighting matrix Ω in (43) is just to enhance the feasibility of related linear matrix inequalities [109]. If Ω_1 and Ω_2 are two design parameters, then for any threshold $\varepsilon \in (0, 1)$, the ETS may be reduced to a time-triggering scheme since there may exist Ω_1 and Ω_2 such that no event will be triggered. Thus, the event-triggering condition (45) works well unless Ω_1 and Ω_2 are given *a priori*. Moreover, from some simulation results, we find that the ETS indeed can be regarded as a time-triggering scheme if we set $\Omega_1 = \Omega_2^{-1}$. However, this finding has not been proven from theoretical points.

The ETSs for both event-triggered sampling and event-triggered transmission mentioned above are designed based on a constant threshold parameter σ or ε . As a matter of fact, the choice of such a threshold parameter greatly affects a data transmission rate, which is in general time-varying, through practical communication channels. For example, raw transmission rates over an IEEE 802.11b wireless local area network may vary over time at 1, 2, 5.5, and 11 Mb/s due to time-varying interference and random wireless fading [110]. It is thus reasonable to consider an essentially time-varying threshold parameter within ETSs. In [110], by assuming that the threshold parameter in the ETS is time-varying with attainable upper and lower bounds, the problem of distributed event-triggered H_∞ consensus filtering for a discrete-time linear system over a wireless sensor network is addressed through a threshold-parameter-dependent approach. Although such an ETS provides certain freedom to adjust the inter-event times, it does not fully explore the relationship between the time-varying threshold parameter and the variant transmission

rates over wireless transmission channels, which leaves much room for improvement. Recently, in the context of multi-agent coordination control, for the i th agent, a dynamical ETS is proposed as [111]

$$\psi_i^T(t_k^i, v) \Omega_1 \psi_i(t_k^i, v) \geq \varepsilon((t_k^i + v)h) x_i^T(t_k^i h) \Omega_2 x_i(t_k^i h) \quad (46)$$

where $\psi_i(t_k^i, v) = x_i((t_k^i + v)h) - x_i(t_k^i h)$ and $\varepsilon(\cdot)$ is a time-varying threshold parameter whose value is adjusted according to a dynamic rule. A key property of the dynamic threshold parameter lies in that $\varepsilon(\cdot)$ is a monotone nonincreasing sequence, which is shown to be beneficial in ensuring a better tradeoff between reducing inter-agent communication frequency and preserving an expected control performance compared with the static ETSs. This is because the static ETSs may not trigger and transmit those data packets that contain useful information to preserve the desired control performance. However, how to evaluate which data packet possesses useful information for guaranteeing system performance is not discussed in [111] and deserves further investigation.

VI. SECURITY CONTROL

The ever-increasing demand for guaranteeing reliable and safe monitoring and operation of modern critical infrastructure, such as smart grids, water and gas supply distribution systems, intelligent transportation systems and so on, has stimulated intensive research and industry interest in developing security control strategies for NCSs.

In this section, we first describe a general security control problem for an NCS with a discrete-time linear plant under various attacks. Then, we classify and discuss some recent results on security control based on two broad categories of data availability and integrity attacks.

A. Security Control Problem

A typical architecture of an NCS is composed of a physical layer and a cyber layer. The physical layer consists of a physical plant and a number of physical components like sensors and actuators, while the cyber layer is comprised of several communication devices, monitors, detectors, controllers and other functions, as shown in Fig. 4.

Suppose that the physical plant is modeled as a discrete-time linear time-invariant system described by

$$\begin{cases} x(k+1) = Ax(k) + Bu(k) + B_w w(k) + B_p a_p(k) \\ y(k) = Cx(k) + Dv(k) + D_s a_s(k) \end{cases} \quad (47)$$

where $x(k)$, $u(k)$, $w(k)$ and $y(k)$ are the system state, control input, external disturbance input (or process noise), and system measurement output, respectively; $v(k)$ is the sensor measurement/communication noise; $a_p(k) = \text{col}\{a_x(k), a_u(k)\}$ denotes the *system attack vector* with $a_x(k)$ and $a_u(k)$ related to the system state and the control input, respectively; $a_s(k)$ is the *sensor attack vector*; and A, B, B_w, B_p, C, D , and D_s are real matrices of compatible dimensions. The system (47) is considered to be in normal operation if $a_p(k) \equiv 0$ and $a_s(k) \equiv 0$ for $\forall k \in \mathbb{N}$.

Sophisticated attackers usually impose various types of attacks upon remote control systems. The malicious impact of an attack on NCSs may be eventually revealed by three key

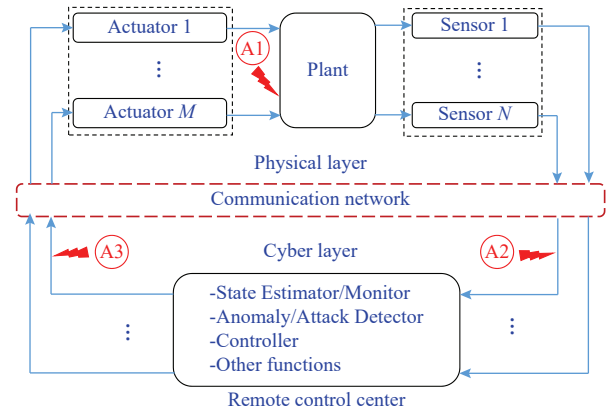


Fig. 4. A general architecture of an NCS under data availability and integrity attacks. A1: data integrity attacks on the system state; A2: data availability and/or integrity attacks on sensor data; A3: data availability and/or integrity attacks on control data.

properties of sensor and control data, i.e., *confidentiality*, *availability*, and *integrity*. Notice that information-theoretic studies on attacks focus mainly on employing some information assurance techniques (e.g., data authentication, access control, cryptographic algorithms) to preserve data confidentiality, while control-theoretic strategies against attacks are devised primarily from evaluating and maintaining the availability and trustworthiness of sensor and control data. This is because information security techniques do not exploit the compatibility of data with the dynamics of the underlying physical plant and they are ineffective against insider or sophisticated attackers targeting dynamics of the physical plant [112]. Moreover, data availability and data integrity are two common security goals in the context of system monitoring and control. From Fig. 4, one can see that the adversary may inject malicious attack signals $a_p(k)$ and $a_s(k)$ on the sensor measurement and/or control commands by compromising their availability and/or integrity.

An attacker's strategy generally relies on three factors: *the attack duration* $\mathcal{T}_a := \{k_0, k_0 + 1, \dots, k_L\}$, *the attack profiles* $\mathcal{F}_a := \{\Theta_1, \Theta_2, \dots, \Theta_L\}$ and *the stealthiness*, where k_0 denotes the attack start time; $k_L = k_0 + L - 1$ represents the attack stop time, and $\Theta_1, \Theta_2, \dots, \Theta_L$ are some compatible non-zero vectors. The attack duration and profiles characterize the adversary's destructiveness while the stealthiness reflects the detectability of the adversary by any anomaly/attack detector. It is noteworthy that the stealthiness of an adversary often depends on *a priori* model knowledge of the plant, disclosure resources and disruption capabilities [22], based on which, the adversary can perform his/her malicious actions by carefully designing the attack vector $a(k) = \text{col}\{a_p(k), a_s(k)\}$ as

$$a(k) = \begin{cases} \Theta_{k-k_0+1}, & k \in \mathcal{T}_a \\ 0, & k \notin \mathcal{T}_a. \end{cases} \quad (48)$$

The attack vector $a(k)$ can be used to reveal the accessibility and integrity of control and sensor data under attack. For example, by properly choosing $B_p a_p(k) = -Bu(k)$ and $D_s a_s(k) = -Cx(k)$, control and sensor data can be blocked on the receiver's side, leading to data interruption; by letting

$a_p(k)$ and $a_s(k)$ be arbitrary non-zero vectors, the integrity of control and sensor data can be compromised. Note that a sophisticated adversary will coordinate the system attack vector $a_p(k)$ and the sensor attack vector $a_s(k)$ to completely manipulate the remote control system without being detected/recovered or with vastly increased time and cost for detection/recovery.

From the analysis above, a security control problem of the system (47) can be stated as: Given the available sensor measurement $\{y(k), k \in \mathbb{N}\}$ of system (47) subject to the attack vector $a(k)$, develop a desirable security control scheme $u(k) = C(y(k))$ such that: 1) the true system state can be estimated accurately and reliably; and 2) the manipulated system can be recovered from attacked operation back to normal. In what follows, some results addressing such a security control issue of NCSs under data availability and data integrity attacks are discussed.

B. Security Control Under Data Available Attacks

Data availability refers to the ability of sensor and control data of being accessible at their destinations. A lack of availability results in an interruption, also called as a denial of service (DoS), of the data. To launch a data availability attack (often occurs at the attack points A2 and A3 in Fig. 4), the adversary can jam the communication channels, attack the routing protocols, flood with network traffic some devices, compromise system components and prevent them from sending/receiving data, etc. Thus, data availability attacks are also referred to as *DoS attacks* [21], [23], [113]–[116] or *jamming attacks* [118].

A direct consequence of a data availability attack on the NCS is that the sensor and control data cannot arrive at their destinations on time, and thus are deemed as “dropped” or “lost”. A common model for such an attack is to choose $B_p a_p(k) = -B_u u(k)$ and $D_s a_s(k) = -B_y C x(k)$, where B_u and B_y are two binary incidence matrices indicating which channels suffer the data interruption. On the other hand, given limited energy budget and/or evitable exposure of himself/herself, an adversary will not persistently but randomly decide when and where to launch the attacks. Furthermore, the attacked data launched on the communication channels may also experience random network congestion and transmission rate constraints, etc. Thus, it seems natural to mathematically model a data availability attack as random packet dropouts, e.g., a Bernoulli-type adversary $B_p a_p(k) = -\alpha(k) B_u u(k)$ and $D_s a_s(k) = -\beta(k) B_y C x(k)$, where $\alpha(k)$ and $\beta(k)$ are Bernoulli random variables with certain probabilities. Several methods to deal with Bernoulli packet dropouts [119], [120] and Markovian packet dropouts [121], [122] can be employed to evaluate the impact of such attacks on the closed-loop system performance, see, e.g., [21], [22] on Bernoulli-type adversaries and [113], [123] on Markovian-type adversaries. It should be also mentioned that packet dropouts caused by data availability attacks need not follow some specific probability distributions. In this sense, new methods on analysis and control design against attacks are proposed at the right moment. To mention a few, in [114], the event-triggered H_∞

load frequency control problem for multi-area power systems under energy-limited DoS attacks is studied, where the maximum DoS attack duration is given. In [21], the optimal control issue on DoS attacks under a maximum number of jamming actions within a prescribed control horizon is considered. The problem of event-triggered control is addressed [115] for NCSs under some “well-structured” pulse-width modulated DoS jamming attacks. To remove the assumption on the “structure” of the DoS attack signal, an output-feedback dynamic event-triggered control framework is presented [23] to cope with a general class of DoS attacks characterized by frequency and duration. It is shown that both resource-awareness and resilience can be investigated based on this unified framework. The above secure control methods are devised just for some certain categories of DoS attacks. Another alternative method is to use the game theory, where the controller and adversary are deemed as two players with competing goals in a game. Taking [116] for example, a game between the defender and the attacker is described as: the defender compensates for the attack-induced performance loss while the attacker drives the control system out of the safety zone using the lowest attacking intensity. Based on the game theory, a number of optimal algorithms for both cyber defenders and DoS attackers are well designed.

Although a data availability attack is not stealthy to system operators, its detection is generally nontrivial because the absence of sensor and control data may be misdiagnosed as network-induced constraints like packet dropouts, which deserves further investigation. In fact, an adversary may purposely take full advantage of randomness to imitate packet dropouts that occur due to congestion or channel constraints [123]. It should be also noted that some existing approaches treat DoS/jamming attacks as packet dropouts when establishing mathematical attack models such that quantitative analysis can be made on performance degradation caused by attacks. How to expose the active behaviors of attackers and the passive impacts of packet dropouts should be further exploited.

C. Security Control Under Data Integrity Attacks

Data integrity refers to the trustworthiness of sensor and control data. A lack of integrity results in deception, false data, or misleading actions. To launch a data integrity attack (often occurs at the attack points A2 and A3 in Fig. 4), an adversary can modify the contents of sensor measurement and control commands by inserting erroneous or misleading data when they are transmitted via communication channels. Note that a data integrity attack is generally more cunning and difficult to tackle as opposed to a data availability attack due to the fact that the tampered data are disseminated in an epidemic manner over the network and may cause the correlated negative impact on the system performance and operation. From the controller's perspective, intensive research efforts have been made on a number of issues: (i) How to accurately and resiliently estimate the true system state when the system is manipulated by data integrity attacks; (ii) How to detect data integrity attacks; and (iii) How to safely control the system when it is under data integrity attacks. Up until

now, various types of data integrity attacks have been identified in the control-theoretic literature, such as *deception attacks* [38], [117], [124]–[126]; *false data injection attacks* [127], [128]; *replay attacks* [20], [22], [129]; *sparse sensor attacks* [130].

The issue (i) is related to the resilient and secure estimation. With the assumption that the data integrity attack signals are bounded [131]–[133] or follow specific probabilistic distributions [38], [126], one of approaches to solving the problem is to model the data integrity attack signals as system external disturbance, process or measurement noise, as done in the conventional literature on robust control and filtering. It should be pointed out that if a naive adversary only makes small perturbations in system dynamics and/or data, such an ineffective attack strategy can be readily combated by the inherently robust and resilient estimators and controllers. On the other hand, significant perturbations caused by the adversary can be easily diagnosed by some anomaly detectors. There is no doubt that a cunning adversary will carefully use the available resources to generate proper perturbations while maximizing the damage to the system.

Notice that the ultimate goal of malicious attacks is to deteriorate the desired system performance or destabilize the underlying system. Designing robust and resilient state estimators is an effective way to mitigate the impact of these attacks on the system performance and operation. Nevertheless, it is essentially desirable to develop suitable detection mechanisms that can locate and identify the attacks in a timely fashion [25], and further warn the system operator to allow corrective actions, thus mapping the issue (ii) into the attack detection. Generally, the following two attack detection approaches have been widely employed in the context of secure estimation and control: 1) χ^2 -detector based on Kalman filters; and 2) Fault detection and isolation techniques. However, such a χ^2 -detector approach requires that the attack signals obey Gaussian distribution [132] and may fail to detect sophisticated deception attacks with good knowledge of the system. The widely-used fault detection and isolation tools, on the other hand, treat attacks and faults in the same manner, which greatly simplifies the detection procedures [134]. When system faults and malicious attacks are both present, this approach might be efficient if one could ascertain the similarities between them. Nevertheless, faults and attacks also possess inherently distinct features in terms of *stealthiness*, *destructiveness*, and *occurrence places*, which make these tools invalid to tackle a smart and cunning attacker. To the best of the authors' knowledge, it remains challenging to establish a rigorous theoretical secure estimation and control framework to effectively identify the effects caused by malicious attacks from faults.

Both issues (i) and (ii) provide the first step towards the security guarantee of the closed-loop system, while issue (iii) represents the ultimate goal of the system designers, preserving the security of the closed-loop system. It is noted that most researches along this line focus on investigating system vulnerabilities or preserving system resilience and safety against a particular type of attacks or some detectable attacks. Nevertheless, it is not uncommon that sophisticated

adversaries may possess sufficient system model knowledge, disclosure and disruption resources to perform *covert (stealthy) attacks* [135]–[137] that can completely bypass traditional anomaly detectors. For example, a typical covert attack model introduced in [135] has the following form [138]

- (*System attack vector*). Depending on the available resources of the adversary, the vector $a_p(k)$ in (47) can be arbitrarily selected;

- (*Sensor attack vector*). The vector $a_s(k)$ in (47) is properly designed as

$$\begin{cases} x_a(k+1) = Ax_a(k) + Ba_p(k), \\ a_s(k) = -Cx_a(k) - B_p a_p(k) \\ x_a(k) = 0, \quad k \leq k_0 \end{cases} \quad (49)$$

where $x_a(k) = x(k) - x_h(k)$ denotes the system state component induced by the attack and $x_h(k)$ is the system state component under no attack.

The principle of the above covert attack strategy lies in the following two aspects. First, the adversary applies some pre-defined data integrity attacks modeled by $a_p(k)$ to corrupt the system state. Second, he or she adds some spurious or misleading data modeled by $a_s(k)$ to cover the changes of system measurement so as to render the attacks completely undetectable and stealthy to the remote detector. A covert attack is completely stealthy to traditional anomaly detectors if system measurement from all sensors are compromised (i.e., $D_s = I$) [135], [138], and can be implemented by using different strategies, such as replaying, false data injecting, zero-dynamics and so on, see [22] in detail.

VII. CONCLUSION AND SOME CHALLENGING ISSUES

NCSs have been a hot topic for over two decades. Boosted by the rapid development of network communication technologies, NCSs have found ever-increasing applications in a wide range of areas, such as smart grids, intelligent transportation, unmanned space (surface/marine) vehicles, robotic networks, and so on [139]. Those applications in turn have stimulated theoretical research of NCSs to higher levels by bringing a large number of practical issues from two primary topics ‘*control of networks*’ and ‘*control over networks*’. This paper has surveyed recent theoretical developments of five control issues from the perspective of ‘*control over network*’, including sampled-data control, quantization control, networked control, event-triggered control, and security control. In what follows, we present a number of challenging issues worthy of future research.

- 1) A co-design scheme is expected for an NCS by simultaneously taking ‘*control of networks*’ and ‘*control over networks*’ into account. With such a scheme, on the one hand, suitable communication protocols [53], [140] can be designed to ensure good QoS of communication networks, and on the other hand, notable control strategies can be provided such that the NCS achieves desirable control performance to carry out expected control tasks;

- 2) In most of event-triggered control results, it is assumed that all event-triggered packets have been successfully transmitted. However, in the open network environments, there may exist cyber attacks, some triggered packets may be

not transmitted to the following points safely and timely. How to develop resilient and secure event-triggered communication schemes in the presence of malicious attacks is an attractive topic;

3) Although a large number of results have been reported on coordination of networked MASs from various perspectives, there remains much room to explore secure coordination of networked MASs subject to various network-induced constraints and attacks simultaneously;

4) A smart grid which incorporates physical power grids, information and communication systems, and human societies is a typical example of networked industrial control systems. Due to the use of communication networks, reliability and security of smart grids may inevitably be threaten and compromised by network-induced constraints and attacks. As a result, it is essential and critical to develop an advanced control method for smart grids that can deliver reliability and security of energy supply, and efficiency of operation by taking network-induced constraints and attacks into account;

5) The majority of existing control literature focuses on ensuring asymptotic or exponential convergence of NCSs under network-induced constraints. In real control implementation, however, it is more desirable and practical for an NCS to achieve fast convergence while preserving good robustness and control performance. It is well acknowledged that adopting fixed-time control can guarantee that the closed-loop systems achieve stability or stabilization in a fixed or prescribed time [141]–[143]. However, it becomes more difficult and challenging for NCSs to ensure the fixed-time convergence due to the presence of aforementioned various network-induced constraints and security threats, which deserves deep investigation; and

6) Besides data availability and data integrity, another concern on security control in NCSs is data confidentiality, which is closely related to privacy protection. How to devise effective privacy-preserving security control schemes for NCSs in the presence of network-induced constraints will be a significant and challenging issue. Furthermore, it will be promising to deeply investigate the impacts of combinational data availability, integrity and confidentiality attacks on control performance of NCSs.

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Xian-Ming Zhang (M’16-SM’18) received the M.Sc. degree in applied mathematics and the Ph.D. degree in control theory and engineering from Central South University, Changsha, China, in 1992 and 2006, respectively.

In 1992, he joined Central South University, where he was an Associate Professor with the School of Mathematics and Statistics. From 2007 to 2014, he was a Post-Doctoral Research Fellow and a Lecturer with the School of Engineering and Technology, Central Queensland University, Rockhampton, QLD, Australia. From 2014 to 2016, he was a Lecturer with the Griffith School of Engineering, Griffith University, Gold Coast, QLD, Australia. In 2016, he joined the Swinburne University of Technology, Melbourne, VIC, Australia, where he is currently a Senior Lecturer with the School of Software and Electrical Engineering. His current research interests include H-infinity filtering, event-triggered control systems, networked control systems, neural networks, distributed systems, and time-delay systems.

Dr. Zhang was a recipient of second National Natural Science Award in China in 2013, and first Hunan Provincial Natural Science Award in Hunan Province in China in 2011, both jointly with Prof. M. Wu and Prof. Y. He, and the IET Premium Award in 2016, jointly with Prof. Q.-L. Han. He is an Associate Editor of the *Journal of the Franklin Institute*, *International Journal of Control, Automation, and Systems* and the *Neurocomputing*, and he is an Editorial Board Member of *Neural Computing and Applications*.



Qing-Long Han (M’09-SM’13-F’19) received the B.Sc. degree in Mathematics from Shandong Normal University, Jinan, China, in 1983, and the M.Sc. and Ph.D. degrees in Control Engineering and Electrical Engineering from East China University of Science and Technology, Shanghai, China, in 1992 and 1997, respectively.

From September 1997 to December 1998, he was a Post-doctoral Research Fellow with the Laboratoire d’Automatique et d’Informatique Industrielle (currently, Laboratoire d’Informatique et d’Automatique pour les Systèmes), École Supérieure d’Ingénieurs de Poitiers (currently, École Nationale Supérieure d’Ingénieurs de Poitiers), Université de Poitiers, France. From January 1999 to August 2001, he was a Research Assistant Professor with the Department of Mechanical and Industrial Engineering at Southern Illinois University at Edwardsville, USA. From September 2001 to December 2014,

he was Laureate Professor, an Associate Dean (Research and Innovation) with the Higher Education Division, and the Founding Director of the Centre for Intelligent and Networked Systems at Central Queensland University, Australia. From December 2014 to May 2016, he was Deputy Dean (Research), with the Griffith Sciences, and a Professor with the Griffith School of Engineering, Griffith University, Australia. In May 2016, he joined Swinburne University of Technology, Australia, where he is currently Pro Vice-Chancellor (Research Quality) and a Distinguished Professor. In March 2010, he was appointed Chang Jiang (Yangtze River) Scholar Chair Professor by Ministry of Education, China. His research interests include networked control systems, multi-agent systems, time-delay systems, complex dynamical systems and neural networks.

Professor Han is one of The World's Most Influential Scientific Minds: 2014-2016, and 2018. He is a Highly Cited Researcher according to Clarivate Analytics (formerly Thomson Reuters). He is a Fellow of The Institution of Engineers Australia. He is an Associate Editor of several international journals, including the *IEEE Transactions on Cybernetics*, the *IEEE Transactions on Industrial Electronics*, the *IEEE Transactions on Industrial Informatics*, *IEEE Industrial Electronics Magazine*, the *IEEE/CAA Journal of Automatica Sinica*, *Control Engineering Practice*, and *Information Sciences*.



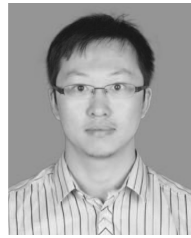
Xiaohua Ge (M'18) received the B.Eng. degree in electronics and information engineering from Nanchang Hangkong University, Nanchang, China, in 2008, the M.Eng. degree in control theory and control engineering from Hangzhou Dianzi University, Hangzhou, China, in 2011, and the Ph.D. degree in computer engineering from Central Queensland University, Rockhampton, QLD, Australia, in 2014.

From 2011 to 2013, he was a Research Assistant with the Centre for Intelligent and Networked Systems, Central Queensland University, Rockhampton, QLD, Australia, where he was a Research Fellow in 2014. From 2015 to 2017, he was a Research Fellow with the Griffith School of Engineering, Griffith University, Gold Coast, QLD, Australia. He is currently a Lecturer with the School of Software and Electrical Engineering, Swinburne University of Technology, Melbourne, VIC, Australia. His research interests include distributed estimation over sensor networks, distributed coordination in multi-agent systems, security and privacy preserving in cyber-physical systems.



Derui Ding (M'16) received both the B.Sc. degree in Industry Engineering in 2004 and the M.Sc. degree in Detection Technology and Automation Equipment in 2007 from Anhui Polytechnic University, Wuhu, China, and the Ph.D. degree in Control Theory and Control Engineering in 2014 from Donghua University, Shanghai, China. From July 2007 to December 2014, he was a teaching assistant and then a lecturer in the Department of Mathematics, Anhui Polytechnic University, Wuhu, China. He is currently a senior research fellow with the School of Software and Electrical Engineering, Swinburne University of Technology, Melbourne, Australia. From June 2012 to September 2012, he was a research assistant in the Department of Mechanical Engineering, the University of Hong Kong, Hong Kong. From March 2013 to March 2014, he was a visiting scholar in the Department of Information Systems and Computing, Brunel University London, UK. His research interests include nonlinear stochastic control and filtering, as well as multi-agent systems and sensor networks. He has published around 40 papers in refereed international journals. Dr. Ding is serving as an Associate Editor

for Neurocomputing. He is also a very active reviewer for many international journals.



Lei Ding (M'18) received the B.Sc. degree in automation, and the M.Sc. and Ph.D. degrees in control theory and control engineering from Dalian Maritime University, Dalian, China, in 2007, 2009 and 2014, respectively. From November 2010 to November 2012, he was a visiting Ph.D. student sponsored by China Scholarship Council at the Centre for Intelligent and Networked Systems, Central Queensland University, Australia. From January 2015 to January 2016, he was a visiting Research Fellow at Western

Sydney University, Australia. From May 2016 to January 2017, he was a Postdoctoral Associate at Wayne State University, USA. From February 2017 to March 2019, He worked as a Research Fellow at Swinburne University of Technology, Australia. In March 2019, he joined Nanjing University of Posts and Telecommunications, where he is currently a Professor with the Institute of Advanced Technology. His research interests include event-triggered control and distributed cooperative control of multi-agent systems and their applications to smart grids.



Dong Yue (SM'08) received the Ph.D. degree in Control Science and Engineering from the South China University of Technology, Guangzhou, China, in 1995. He is currently a Changjiang Professor and Dean of Institute of Advanced Technology, Nanjing University of Posts and Telecommunications. His research interests include analysis and synthesis of networked control systems, multi-agent systems, optimal control of power systems, and internet of things.

He is currently an Associate Editor of the *IEEE Transactions on Industrial Informatics*, the *IEEE Transactions on Neural Networks and Learning Systems*, *Journal of The Franklin Institute*, *International Journal of Systems Science* and the *IEEE Control Systems Society Conference Editorial Board*.



Chen Peng (M'13–SM'15) received the Ph.D. degree in control theory and control engineering from the Chinese University of Mining Technology, Xuzhou, China, in 2002. From 2004 to 2005, he was a Research Associate with the University of Hong Kong, Hong Kong. From 2006 to 2007, he was a Visiting Scholar with the Queensland University of Technology, Brisbane, QLD, Australia. From 2011 to 2012, he was a Post-Doctoral Research Fellow with the Central Queensland University, Rockhampton, QLD, Australia. In 2012, he was appointed as an Eastern Scholar by the Municipal Commission of Education, Shanghai, China, and joined Shanghai University, Shanghai, where he is currently the Director of Centre of Networked Control Systems and a Distinguished Professor. In 2018, he was appointed as an Outstanding Academic Leader by the Municipal Commission of Science and Technology, Shanghai. His current research interests include networked control systems, distributed control systems, security control, and intelligent control systems. Prof. Peng was a recipient of the one of the Most Cited Chinese Researchers Award in Computer Science by Elsevier from 2014 to 2018. He is an Associate Editor of a number of international journals, including *Information Sciences* and *Transactions of the Institute of Measurement and Control*.

He is currently an Associate Editor of the *IEEE Transactions on Industrial Informatics*, the *IEEE Transactions on Neural Networks and Learning Systems*, *Journal of The Franklin Institute*, *International Journal of Systems Science* and the *IEEE Control Systems Society Conference Editorial Board*.